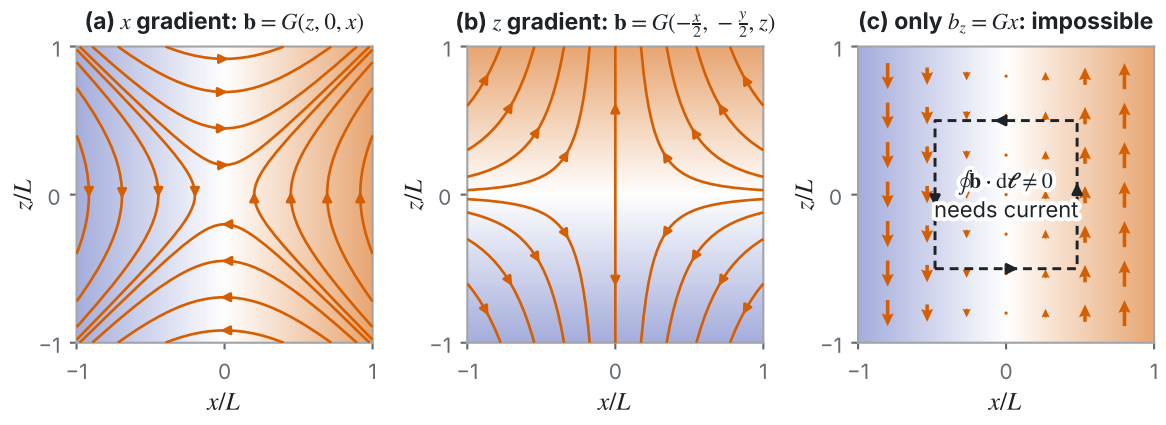
(a) x gradient: $\mathbf{b} = G(z, 0, x)$
(b) z gradient: $\mathbf{b} = G(-\frac{x}{2}, -\frac{y}{2}, z)$
(c) only $b_z = Gx$: impossible
$\oint \mathbf{b} \cdot d\boldsymbol{\ell} \neq 0$
needs current
z/L
x/L
1
0
−1